

Roll No.

TCS-591

**B. TECH. (CSE & CE SPL) (FIFTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

COMPUTER SYSTEM SECURITY

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Why do we need access control process ? Explain the procedure of role-based access control. Use figure in your answer. (CO1)

OR

- (b) Why do we need a robust application software. What are the different tools and techniques for writing the robust application software ? (CO1)
2. (a) How would you compare "static vulnerability analysis and dynamic vulnerability analysis" ? Provide the details of fault injection and dynamic taint methods of vulnerability analysis. (CO2)

OR

- (b) Demonstrate the procedure of buffer overflow attack using a programming example. Discuss any *two* methods, which can be used to prevent the buffer overflow attack. (CO2)

P. T. O.

(2)

3. (a) Discuss the procedure of shellshock attack. What are the devastating effects of shellshock attack ? (CO3)

OR

- (b) Investigate the procedure of fuzzing using a figure. What are the issues with fuzzing ? (CO3)

4. (a) Can a format string vulnerability be exploited ? If so, explain the procedure through a programming example. (CO4)

OR

- (b) Demonstrate the working mechanism of heartbleed attack. How can we mitigate the heartbleed attack ? (CO4)

5. (a) How would you compare "authentication, access control and authorization mechanisms" ? (CO5)

OR

- (b) Write down the differences between sandboxing and isolation. What are the different ways of implementation of a sandbox ? (CO5)